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MILITARY STANDARD

FUZE, DESIGN SAFETY, CRITERIA FOR

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20301

Design, Safety, Fuze, Criteria For
MIL-STD-1316A

1. This Military Standard is mandatory for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions or deletions should be addressed to the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-ND7.

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FOREWORD

The purpose of this standard is to establish specific design safety requirements and objectives for fuzes (to include safety and arming devices) intended for use with munitions. It is the policy of the Department of Defense that safety and arming requirements, objectives, and design guides specified herein are mandatory fundamental elements of design, engineering, production and procurement of fuzes. Fuzes shall provide safety which is consistent with operational use, handling, assembly, storage, transportation and operational readiness.

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1. SCOPE

1.1 This standard establishes the design safety criteria for fuzes (to include safety and arming (S&A) devices).

1.2 This standard applies to fuzes and S&A devices as follows:

a. Fuze exploratory developments to meet future requirements where such developmental work has not progressed beyond the conceptual phase, and where work has progressed to the point of assignment of an experimental model designation.

b. Fuze developments in progress when directed and to the extent directed by appropriate authority, to include those fuzes in:

1. Advanced development;
2. Engineering and operational system development; and
3. Advance production engineering (APE).

c. This standard applies to fuzes and S&A's for the following munitions:

1. Guided Missile Warheads
2. Rocket Warheads
3. Projectiles
4. Bombs
5. Bomblets
6. Destruct Systems
7. Cluster Bomb Dispensers, Free Fall

1.3 This standard does not apply to fuzes which were type classified as "Standard" prior to the effective date of this standard, or to fuzes for the following munitions:

a. Army Missile Systems currently in the Engineering Development (ED) phase or beyond;

b. Nuclear Munitions;

c. All types of grenades;

d. Munitions dispensed or emplaced manually such as mines, flares, signals, markers, demolition items, and other special warfare items.

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2. REFERENCED DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-339	- Tetryl (Trinitrophenylmethylnitramine)
MIL-R-398	- RDX
MIL-E-14970	- Explosive Composition A5
MIL-R-21723	- RDX Composition CH-6
MIL-I-23659	- Initiators, Electric, Design and Evaluation of
MIL-P-24014	- Preclusion of Hazards from Electromagnetic Radiation to Ordnance, General Requirements for
MIL-P-45486	- Pellets, RDX
MIL-P-46464	- Pellets, Tetryl
MIL-E-81111	- Explosive, Plastic-Bonded Molding Powder (PBXN-5)

NAVY WEAPON SPECIFICATIONS

WS-4660	- Dipam Explosive
WS-5003	- HNS, Explosive

STANDARDS

MILITARY

MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-444	- Nomenclature and Definitions in the Ammunition Area
MIL-STD-882	- System Safety Program For Systems and Associated Subsystems and Equipment Requirements For
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment and Facilities

OTHER PUBLICATIONS

NOLC Report 666 - Experimental Methods For Determining the Effectiveness of Interruption of a Fuze Explosive Train, 1 July 1966

AFR 136-8 - Responsibilities of Explosive Ordnance Disposal

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3. DEFINITIONS

3.1 The definitions of MIL-STD-444 apply to the ammunition terms used in this standard. For interpretation of this standard, the following specific definitions apply:

3.1.1 Arm.-To make a fuze ready for functioning by removal of the firing train interrupter, by removal of safety devices, or by alignment of the explosive elements in the explosive train of the fuze.

3.1.2 Arming Delay.-A time delay between launch initiation and fuze arming employed to obtain a safe separation distance.

3.1.3 Assembled Fuze.-The completed fuze with all component parts put together; a fuze requiring no added components or parts to prepare it for installation into the round (projectile, rocket, bomb, etc.) in which it is to function. Assembling the fuze is the process of putting the parts and components together.

3.1.4 Design Guides.-Concepts, logic, background, examples, statements of good practices or rules, time-proven concepts or features, any of which can be used as reminders to aid the thought process of the designer in his effort to obtain optimum safety through design.

3.1.5 Dud.-A malfunction of a munition which results in failure to explode or function, although functioning was intended.

3.1.6 Environment.-The total set of physical conditions to which the fuze is exposed.

3.1.7 Environmental Force.-A stimulus obtained from the environment.

3.1.8 Early Burst (Prior Function).-A malfunctioning in which a munition explodes or functions after safe separation but before the expected time or circumstance. For proximity fuzes following safe separation, an explosion prior to design Height of Burst (HOB) environment is considered an early.

3.1.9 Fail Safe Feature.-A characteristic of a fuze system or part thereof which is designed to result in a dud round when safety feature(s) malfunction.

3.1.10 Fuze Installation.-The act of installing or inserting the assembled fuze into the round (projectile, rocket, bomb, etc.) in which it is to function; sometimes called "fuzing", e.g., fuzing the bomb.

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3.1.11 Fuze System.-A complete system including fuze, fuze setter, system interfaces (electrical and mechanical), and any other ancillary devices necessary for fuze operation.

3.1.12 Fuze Safety System.-The aggregate of devices (environment sensors, launch event sensors, command functioned devices, logic networks) included in the fuze to provide safety, and which prohibits arming and functioning of the fuze through all environments during transportation, storage, handling, installation, and launching of the munition until a safe separation distance has been achieved.

3.1.13 Independent Safety Feature.-Each of several features is defined as independent if the integrity of the safety feature is not affected by the function or malfunction of one or more other safety features, or by sensing the same environment utilized by one of the other safety features.

3.1.14 Launch Cycle.-The period between the time the munition is irreversibly committed to launch and the time the munition leaves the launcher.

3.1.15 Main Charge.-The charge which is provided to accomplish the end result in a munition, i.e., bursting a casing to produce blast and fragmentation, splitting a canister to dispense sub-missiles or producing other effects for which it may be designed.

3.1.16 Objective.-A goal for Fuze development which should be incorporated in a fuze design or concept when advancement in the state-of-the-art indicates that attaining the objective is reasonable.

3.1.17 Premature Burst.-A malfunctioning in which a munition explodes or functions before safe separation.

3.1.18 Requirement.-A feature or procedure which must be a part of the fuze or fuze development.

3.1.19 Sensor, Sensing Elements.-A component or series of components designed to detect and react to an environment.

3.1.20 Safe and Arming Device.-A mechanism whose primary function is to prevent an unintended detonation and in turn allow the explosive train of the munition to function after the munition has reached the proper point in time or space.

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3.1.21 Safety System Failure Rate.-The rate is defined as the ratio of safety system failures to the fuzes expended in all phases from munition manufacture up to the attainment of the safe separation distance (or time) which the safing system was designed or required to provide prior to intentional arming or to intended functioning, where an early burst (prior function) is considered an overhead safety problem.

3.1.22 Safe Fuze Design.-A fuze design which limits the probability of death or serious injury to friendly troops, or property damage to an acceptable level.

3.1.23 Safety Feature.-An element or combination of elements which prevent unintentional arming and unintentional detonation.

3.1.24 Safe Separation Distance.-A space relationship between weapon or launcher and munition following launch, beyond which an acceptably low probability of injury to launch or other personnel or damage to the weapon or launcher exists, if the munition should function.

3.1.25 Safety System Failure.-A failure of the fuze system prior to intended detonation which makes the fuze dangerous to handle, launch, or results in a hazardous initiation of one of its explosive components.

3.1.26 Stored Energy.-A component contains stored energy if the component is capable of expending energy in addition to the energy required to initiate its function. Examples of stored energy components are:

a. Cocked Springs
b. Batteries
c. Explosive Components (Bellows and Dimple Motors, etc).

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4. REQUIREMENTS (see 7.4)

4.1 Design.-The following features, procedures, or controls shall be incorporated into the design of fuzes covered under the scope of this document.

4.1.1 Explosive Train Interruption (Tri-Service).-When the explosive train includes primary explosives more sensitive than those listed in Paragraph 4.1.2, at least one interrupter shall separate the primary explosives from the main charge before arming. Explosive train interruption will be positioned so as to isolate primary explosives using at least one interrupter (barrier, shutter, slider, rotor) to separate this explosive from the lead and booster explosive until the fuze arms. The interrupter shall have positive, mechanical locks. Early in design the effectiveness of interruption shall be determined by techniques described in:

a. MIL-STD-331 (Tri-Service)
b. NOLC Report 666 (Navy-Air Force)

4.1.2 Explosive Sensitivity (Lead and Booster Explosives).-The following explosives are the only ones permitted in a position leading to the initiation of the main charge without interruption when the fuze is in the safe condition:

Explosive	Specification
Tetryl	MIL-T-339
RDX Comp CH6	MIL-R-21723
Tetryl, Pellets	MIL-P-46464
DIPAM	WS 466
HNS Type 1 or Type 2 Gr A	WS 5003

4.1.3 Alternate Lead and Booster Explosives.-If it is desired to use explosives other than those listed in 4.1.2, specific written approval is required from the cognizant technical authority (see paragraph 7.4).

4.2 Initiators "in-line".-Initiators "in-line" (i.e., not followed by explosive train interruptions) shall not be used in fuzes, except as allowed by paragraph 4.2.2 below, even though explosives employed are those listed in 4.1.2 Where electrical type initiators or detonators are employed, a positive means (e.g., shorting or switching) of preventing fuze detonation prior to fuze arming shall be provided.

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4.2.1 When the explosive train does not contain primary explosives (e.g., Exploding Bridgewire (EBW) per MIL-I-23659), and has no provision for interruption, shielding, and other protection, then the initiation system shall be designed to provide at least the same degree of fuze safety, including a mechanical interruption in the electrical circuit, as is obtained with an interrupted train employing primary explosives. The effectiveness of this system as a minimum shall be equivalent to that determined by techniques described in:

a. MIL-STD-331 (Tri-Service)
b. NOLC Report 666 (Navy/Air Force)

4.2.2 EBW Devices.-Exploding bridgewire (EBW) devices may be used without subsequent explosive train interruption if the following conditions are met:

a. The explosive initiated by the exploding wire is an explosive listed in 4.1.2.
b. The arming and triggering signals for initiating the EBW device are switched by two independent features requiring independent sources of energy from an environmental force for operation.
c. One of the mechanisms in (b) above shall derive its energy from an environmental force after launch.
d. The sensitivity of the EBW device to electrical initiation is not greater than Group B per MIL-I-23659. The device cannot be initiated by any electrical signal at a peak potential of 500 volts, nor can a 500 volt capacitor discharge, especially from the firing circuit capacitor, initiate the device.

4.2.3 The effectiveness of the circuit interruption provided by the arm and trigger switches will be determined by appropriate tests (note paragraph 7.2) under both normal and abnormal environmental conditions.

4.3 Fuze Safety System

4.3.1 The safety system of fuzes shall comprise at least two independent safety features, each of which is capable of preventing unintentional arming and unintentional detonation of the main charge of the munition. Each of these safety features shall be actuated by at least one separate environmental force. Each safety feature shall employ an independent source of arming energy. One feature shall provide safe separation. One of the mechanisms shall depend upon sensing a post launch environmental

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condition, which unlocks or arms the interrupted train. A launch signal may be used to initiate one of the safety features.

4.3.2 (Army) If the primary explosive material, which is more sensitive than the lead and booster explosives, (see 4.1.2) are housed in the interrupter, a single interrupter locked by the two independent safety features, is acceptable since an omission of the interrupter shall not result in a safety failure. If the element containing the primary explosive is mounted "in-line" and if the omission of a single interrupter could lead to a safety failure, the design shall include two interrupters, each of which shall respond to a different environmental force, or both interrupters shall each respond to two environmental forces.

4.4 Safe-Arm Indication For Bomb Fuzes (Tri-Service).-In addition to those requirements specified in 4.5 below, bomb fuzes must possess a feature(s) which assure a positive means of determining the safe or armed condition of the fuze before and after installation into the bomb.

4.5 Safe-Arming Indication (Navy-Air Force).-One or more of the following options shall be selected or combined in the fuze design:

4.5.1 A feature which assures a positive means of determining safe or armed conditions prior to installation into the round.

4.5.2 A feature which prevents installation of an armed, assembled fuze into the round.

4.5.3 A feature which prevents accidentally assembling the fuze in the armed or partially armed condition. If after installation in the ammunition the fuze could be armed manually, or by any method other than by launching the ammunition, a feature which allows observation or permits ready determination of the safe or armed condition of the fuze after it has been installed in the ammunition must be present.

4.6 Safety System Failure Rate (Tri-Service).-The system safety failure rate of a new fuze shall be determined by performing a Safety Failure Analysis. (A method is described in paragraph 6.3.2).

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5. OBJECTIVES

5.1 Objectives.-The following characteristics, features and controls shall be the objective(s) of safe fuze design. Necessity for deviations from these objectives shall be documented in both detail and summary format showing that compliance is not feasible on the basis of cost, complexity, conformity with other requirements documents, or is beyond the state-of-the-art.

5.2 Energy for Arming.-The fuze safety system should not incorporate a stored energy mechanism which is used to remove an interrupter of the fuze explosive train.

5.3 Fuze systems should incorporate fail safe features based on their applicability to system requirements.

5.4 For multi-stage missiles and rockets, arming mechanisms should utilize last stage initiation as part of the arming sequence.

5.5 Documentation.-The evaluation program used as the basis of the safety statement which is prepared by the developing agency shall be documented in both detail and summary format.

5.6 Quality Assurance.-The design of the fuze should facilitate the fuze connection to inspection and test equipment (in an assembled condition) for all characteristics which assure the safety and intended functioning of the fuze. The fuze design should facilitate the use of automatic inspection equipment to verify the above characteristics, and the design should allow for possible connections with in-flight telemetry test devices.

5.7 Safe-Arm Indication.-One or more of the following options shall be selected or combined in the fuze design:

5.7.1 (Tri-Service).-A positive means of determining the safe or armed condition of the fuze before and after installation into the munition should be employed. This goal applies particularly to fuzes with projecting arming features which could be compromised by tampering.

5.7.2 (Army).-A feature which assures a positive means of determining safe or armed conditions prior to installation into the round.

5.7.3 (Army).-A feature which prevents installation of an armed, assembled fuze into the round.

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5.7.4 (Army).-A feature which prevents accidentally assembling the fuze in the armed or partially armed condition. If after installation in the ammunition the fuze could be armed manually, or by any method other than by launching the ammunition, a feature which allows observation or permits ready determination of the safe or armed condition of the fuze after it has been installed in the ammunition must be present.

5.8 Safety System Failure Rate.-A safety failure rate shall be used as a goal in the design of fuzes. It shall be employed to the extent possible by test or prediction in fuze evaluation. This safety failure rate is the ratio of safety system failures to the fuzes expended in all phases from manufacture up to the designated service end point, either intentional initiation of arming, or to intended functioning. The safety failure rate of bomb fuzes should not exceed the Navy/Air Force goal of one failure in 1×10^6. The safety failure rate of artillery (projectile) fuzes shall not exceed the Army goal of one failure in 2.5×10^6. Safety failure rates for other classes of fuzes have not been specifically defined to date; but as a minimum goal, they should not exceed the one failure in 1×10^6 rate established for bomb fuzes.

5.9 Explosive Ordnance Disposal.-Fuze systems should incorporate Explosive Ordnance Disposal (EOD) features based on the applicability of EOD to system requirements.

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6. DESIGN GUIDES

6.1 For electrical/electronic fuzing of missiles, artillery and mortar projectiles, rockets and bombs, the following guidelines should be utilized:

6.1.1 Arming Signals.-At least two separate and independent functions should be required to arm the fuze. Whenever practicable, the functions should be derived from two independent and dissimilar environments. For example, one signal could be applied to the fuze by actuation of an accelerometer designed to sense a unique acceleration. This would insure that the vehicle (missile, rocket, or projectile) had experienced a satisfactory launching or firing. The second signal could be derived from a device such as a barometer to ascertain that the vehicle had reached an acceptable altitude. Arming devices should be sensitive to different environments so that an incident or accident (e.g., dropping) would not arm the fuze. In the design of some projectiles and some missiles, both spin and setback could be employed to complete arming, since it is unlikely that the shell or missile would experience the two proper forces unless actually fired.

6.1.1.1 Continuous-Arming Signal.-At least one of the signals needed to arm the fuze should be dependent on the continuous application of power. For example, in fuzes which have an arm-safe relay, continuous application of electric power should be required to hold the relay in the armed condition. Loss of power should cause the relay to return to the safe or relaxed position. Thus, vulnerability to spurious pulse-type signals resulting from an accident or electromagnetic interference would be decreased. The probability of an accident causing the generation and application of continuous signals is considered to be less than that of one causing the generation and application of pulse signals. The system design should also provide for the dissipation of stored energy within a reasonable time upon the removal of continuous power.

6.1.1.2 One-Point Arming.-Circuits should be arranged so that a malfunction or application of power to any single point cannot arm the fuze. If all of the arming signals are furnished the fuze by a single circuit, a short to this circuit could conceivably arm the fuze. A condition such as this is referred to as "one-point arming", and is to be avoided.

6.1.1.3 Critical Contacts.-Contacts of cable connectors should be assigned so that those in arming and firing circuits are isolated to avoid critical short circuits. Component input and output circuits should be isolated to prevent bypass. Separate isolated connectors should be used.

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6.1.1.4 Ground Seeking Switches.-Ground seeking switches are those that apply a ground to complete an electrical circuit. This switching scheme should not be employed in arming and firing circuits since the ubiquity of ground provides increased probability of accidentally bypassing the switch by short circuit.

6.1.1.5 Thermal Protection.-Circuits to be energized by thermal batteries should be protected with devices that isolate the battery output when exposed to abnormally high temperatures. These devices should also be considered for thermal protection of circuits using other thermal batteries, since shorts caused by fire may permit application of power to functional circuits.

6.1.1.6 Independent Failure.-Control or protective functions should be designed and arranged to be independent so that if one should fail, an associated control or protective function would not be caused to fail.

6.1.1.7 Arm-Safe Switches.-Arm-safe switches should be capable of being reversed. Electrically driven switches should also be designed so that if armed, they return to the safe state automatically when power is terminated.

6.1.1.8 Safe Plugs.-Safe plugs are used to disenable circuits while fire plugs are used to complete or enable these circuits. A Safe Plug should be provided to interrupt at least one critical fuze arming line, and, in a missile system, the launch command line. Where the latter is not practical, the Safe Plug may apply only to the Fuzed Warhead Section. This Safe Plug can be interchanged with the Fire Plug. The Plug receptacles on the surface of the assembly make functional circuits vulnerable to deliberate unauthorized acts. Therefore, the Safe Plug should be designed to interrupt no more than one of the critical arming functions.

6.1.1.9 In any system, the warhead connector should be externally unavailable when assembled in a warhead section. Where this is not practical, locked covers or similar devices should be employed to restrict access to this connector.

6.1.1.10 Critical components and cabling should be potted so that attempts to gain access will damage the assembly or reveal evidence of tampering.

6.1.1.11 Critical cables and connectors should be positioned so that they are inaccessible.

6.1.2 Manual Operation.-When critical operations requiring human actions must be performed, the design should provide maximum protection against human error. This can be done by limiting

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access to critical points and by minimizing the extent of human actions. Consideration should be given to the following:

6.1.2.1 Electrical connectors should be designed to make improper mating of connectors virtually impossible. Symmetrical arrays of contacts or placement of keys in electrical connectors should be avoided. Pin arrangements should not be used to provide for connector indexing or orientation. Each system should be analyzed to insure that it does not contain connectors with similar keys or contact arrays at locations in the assembly where they may be mismated or interchanged. The design should also provide for ease of mating when connectors are in blind locations. Connector design should preclude misindexing the insert with respect to the shell during assembly by the manufacturer. Designs should provide for maximum protection against electrical faults due to presence of moisture. The probability of mismating can be further reduced by the use of hard connector inserts.

6.1.2.2 Safe Plugs are intended to disable circuits while Fire Plugs are used to complete or activate these same circuits. Both plugs are normally placed in the same receptacle at different times and should be distinctive in appearance so as to reduce the chance for error. Color coding, together with distinctive shapes or sizes, can be used to insure proper identification of these plugs. Different streamers on Safe and Fire Plugs can aid in indicating the status of a weapon. Safe and Fire Plugs for training weapons should be made to be physically incompatible with war reserve weapon receptacles. Contact arrays and keying arrangements for training should be different from war reserve designs. Safe and Fire Plugs should be marked "Safe" or "Fire" as appropriate.

6.1.2.3 Use of switches which require at least two distinct manual actions for operation protects against inadvertent actuation. An example of this characteristic is a rotary option-select switch which must be depressed before being rotated off the safe position. Switch covers and guards may be employed on launching panels to protect against inadvertent switch actuation.

6.1.2.4 Multiple sequential actions should be necessary to launch a guided missile or rocket. Weapon controls and circuits should be designed to prevent firing when improper control operation or improper operating sequence is undertaken. When practicable designs should insure that it is mechanically or electrically impossible to actuate control circuits in improper sequence. Sequential dependent functions can provide safety by being interlocked.

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6.1.6 Static Discharge.-Discharge of static electricity may

6.1.3 Fuze designs may incorporate electrically actuated devices which must be operated remotely. Operating currents may exceed levels normally used for monitoring. Safety degradation can be reduced in these instances by providing for:

6.1.3.1 A shielded cable which accommodates the actuating or control function only and no other. This reduces the probability of energizing other circuits as a result of electrical faults.

6.1.3.2 Isolation of the control portion of the devices from the functional circuits. The actuating or control cables should use connectors separate from functional connectors.

6.1.4 Monitoring.-Components of the fuzed warhead section can be monitored without degrading safety if certain principles are observed as follows:

6.1.4.1 Monitoring current should be limited to the lowest practical value, and must be below the level capable of functioning the most sensitive component of the fuzed warhead section. No monitoring of electro-explosive devices (EED's) is normally allowed aboard ship. If an EED is to be tested, the monitoring current should not exceed ten (10) percent of the maximum No Fire current of the EED.

6.1.4.2 Monitoring currents should not be placed on functional circuits. If monitoring is required, it should be done through separate circuits leading to separate contacts of relays, explosive switches, safing and arming devices, or baroswitches, etc.

6.1.4.3 Monitoring should be accomplished by devices that are inherently limited to the safe allowable current levels.

6.1.4.4 Where power sources that are not inherently current-limited devices are used, malfunctions will not produce currents exceeding the safe allowable maximum.

6.1.5 Component Selection.-Safing components should fail-safe. For example, an arm-safe relay should require an electric current to hold it in the armed position and power removal should cause the relay to return to the safe position. Mechanical initiators for thermal batteries should not activate the battery in case of failure. Component design should also consider all environments to which the system may be exposed, such as shock, fire and electro-magnetic radiation. When possible, high energy initiated detonators and electro-explosive devices should be used.

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for firing squibs must be sufficient to fire them with an

devices, is referred to as predesign analysis of potential hazards. This should be a normal thought process of every designer. Formalizing it in report form is an aid in assuring thoroughness and completeness.

6.3.2 Safety Failure Mode and Effects Analysis.-A useful method for analyzing the safety of a fuze design is a safety failure mode and effects analysis considering potential accident and malfunction paths. This analysis is a systematic consideration of the effects on fuze safety of such things as omission of parts, breaking of parts, malfunctioning of parts, out of order sequence, inspection procedures and personnel errors. It should consider abnormal as well as normal environments, conditions, and personnel actions. The series of events which can lead to an accident (the accident paths) are examined to determine that adequate precautions to preclude them have been taken in design or through inspection and administrative procedures. As in the potential hazards analysis, severe accident situations should be included and judgement and experience must be used to give more emphasis to situations which are more likely to occur. This analysis differs from the predesign analysis of potential hazards in that it is the study of the fuze safety on the basis of known or anticipated hardware characteristics and specifications intended to be incorporated in the hardware design, whereas the predesign analysis is a study to determine and establish desirable safety characteristics and specifications prior to hardware design.

6.3.3 The following guidance is given in regard to design features of the fuze safety system:

The basic fuze safety system should be designed to sense (and require for arming) conditions which are unique to the normal munition launch and post launch cycles that the probablitiy of encountering the same condition at any time other than during a proper launch is negligible.

6.4 Safety Features of Classes of Fuzes.-The best features of proven designs may be considered as guidelines for new designs. In this regard, the following considerations apply to classes of high volume fuzes. The term "classes of fuzes" as used herein refers to broad groupings of fuzes used with weapons whose operational, release, or launch conditions are similar and therefore the forces available for arming during release or launch are similar.

6.4.1 Gun (Tube) Fired Projectile Fuzes

a. Both setback and spin sensors or sensoring elements

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are recommended for use in actuating arming devices in fuzes for projectiles used in rifled guns or weapons.

b. Setback and a bore-rider, or equally effective device, should be used to actuate arming devices in fuzes for projectiles used in smooth bore guns or weapons. (A bore-rider is not considered to be an optimum device).

c. There should be an arming delay for the projectile to clear the muzzle of the gun by a safe separation distance as determined by experience or effects studies.

6.4.2 Aircraft-Carried, Free-Fall Weapon Fuzes

a. Operation of at least two arming devices should not occur until one has sensed or been actuated by intentional separation from the aircraft, including internal stores, and one has sensed or been actuated by a flight or free-fall environment and these events have occurred in the order given.

b. After separation from the aircraft, arming should be delayed until a safe separation is reached.

c. Velocity sensors, if used, should require the minimum air velocity for operation to have been made as far above non-flight conditions as can reasonably be done without endangering operability.

6.4.3 Rocket and Guided Missile Fuzes

a. At least one arming mechanism should be operated by acceleration or motor pressure. Acceleration is the preferred choice.

b. Arming should be delayed by mechanical means until the accumulated acceleration-time or pressure-time signal after integration indicates a safe separation has been reached.

6.4.4 Destructor S and A's.-Two independent signals should be required for the arming of destructor S and A's.

6.4.5 Bomblet Fuzes

a. Actuation of at least one arming device should not occur until it has sensed or been actuated by an operational environment after launch.

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b. Air vanes, if used, should require the minimum air velocity for operation to be as far above nonflight conditions existing external to the dispenser as can reasonably be done without endangering operability.

c. Arming should be delayed until simultaneous actuation of the bomblets could be sustained without disastrous effects to the manned launcher.

d. An additional lock against arming should be provided by the assembly of the bomblet into the cluster or container. This lock would be released by separation or dispersion of the bomblets.

e. The fuze should be designed with a capability of the detonators to fire against a barrier in the out-of-line position in the event they are fired inadvertently or prematurely.

6.4.6 Other Fuzes.-The arming of fuzes other than those previously listed in 6.4 shall be guided by the requirements and objectives applicable to all fuzes and any special provisions directed by the staff of the cognizant Defense agencies.

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7. NOTES

7.1 The restricted list of explosives appearing in Paragraph 4.1.2 of this standard is temporary. Test and acceptance standards are being prepared for Tri-Service qualification of the explosives listed below, and for others for fuzing purposes.

TABLE I

Explosive	Specification	Navy	Air Force	Army
RDX	MIL-R-398*	No**	Yes	Yes
Expl. Comp A5	MIL-E-14970	No	Yes	Yes
RDX, Pellets	MIL-P-45486	No	Yes	Yes
PBXN-5	MIL-E-81111	Yes	Yes	No

* RDX, Type B, Class C or D, per MIL-R-398C with 2% binder lubricant. Binder lubricant is to consist of not more than 25% graphite, and not less than 75% calcium resinate.

** Navy evaluating on the basis of existing data.

7.2 The effectiveness of interruption of the electrical path to the initiator in EBW systems used "in-line", as described in 4.2.2, must be as adequate and positive in security as that provided in explosive train interruption in conventionally initiated systems. Ordinary electrical switches, even though rated for the voltages intended to be used, are not necessarily to be considered adequate. A more complete treatment of this subject will be covered in a supplementary report or standard to be published subsequently.

7.3 The requirements for system safety programs are given in MIL-STD-882, System Safety Program For Systems and Associated Subsystems and Equipment, Requirements For. The system safety criteria developed in accordance with MIL-STD-882 may impose additional requirements or objectives on the fuze design.

7.4 Designs which do not meet the requirements of this standard or do not have specific written approval must be presented to an evaluation board for a safety evaluation and for waiver consideration as follows:

7.4.1 For Army: Commanding General
USA Munitions Command
ATTN: AMSMU-RE
Dover, New Jersey 07801

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7.4.2 For Air Force: Non Nucleur Munitions Safety Group
ATTN: ADTC/DLGS Secretariat
Eglin Air Force Base
Eglin, Florida 32542

7.4.3 For Navy: Commander
US Naval Ordnance Systems Command
ATTN: Code 048
Washington, D. C. 20360

Custodians:
Army-MU

Navy-OS

Air Force-70

Project Number: 1390-0776

Preparing Activity:
Army-MU

Reviewing Activities:

Army-MU, MI
Navy-AS, OS
Air Force-11, 18, 70

NOTE: Certain provisions of this standard are the subject of international standardization agreements ASCC-AIR-STD-20/9 Design Safety Principles For Airborne Weapon Fuzing Systems and STANAG No. 3525 Design Safety Principles For Fuzing Systems. When revision or cancellation of this standard is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

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